

Check Armstrong

</> Check Armstrong



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Problem statement

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You are given an integer '*n*'.

Return '*true*' if '*n*' is an Armstrong number, and '*false*' otherwise.

Note:

An Armstrong number is a number (with '*k*' digits) such that the sum of its digits raised to '*k*'th power is equal to the number itself. For example, 371 is an Armstrong number because $3^3 + 7^3 + 1^3 = 371$.

Example:

Input: '*n*' = 1634

Output: true

Explanation:

1634 is an Armstrong number, as $1^4 + 6^4 + 3^4 + 4^4 = 1634$

Sample Input 1 :

1

Sample Output 1 :

true

Explanation of Sample Input 1 :

1 is an Armstrong number as, $1^1 = 1$.

Sample Input 2 :

103

Sample Output 2 :

false

Expected Time Complexity:

Try to solve this in $O(\log(n))$.

Constraints:

$1 \leq n \leq 10^9$

C++ (g++ 5.4) ▾

```
1 bool checkArmstrong(int n){
2     int last_digit = n % 10;
3     int x = n;
4     int sum = 0;
5     int cnt = (int)(log10(n) + 1);
6     while(x!=0)
7     {
8         last_digit = x%10;
9         sum = sum + pow(last_digit, cnt);
10        x = x/10;
11    }
12    if(sum == n)
13    {
14        return true;
15    }
16    return false;
17 }
18
```

code in c++

Current submission

Report

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Test cases

EXP

Language

Penalty

6/6

80/80

C++

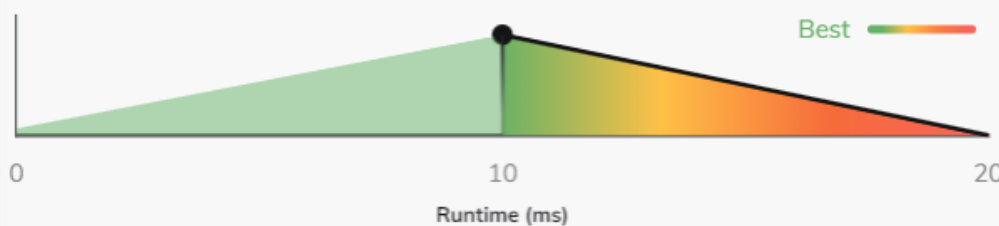
0%

Runtime graph

Memory graph

You are better than 93.62%

Runtime 11 ms



Did you find these test cases useful?



$T.C = O(\log_{10}(n))$

$S.C = O(1)$



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